**Sheet 45** Tue. 2 March 2027 Affine functions: rate of change and variations

**1.** Warm-up · Flash $f(x) = 3x - 1$. Calculate $\dfrac{f(4) - f(1)}{4 - 1}$.

**2.** Core · Flash Direction of variation of $f(x) = -4x + 7$ ?

**3.** Core + · Proof Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.

**4.** Back · Table Ages of the members of a club.

| Age | 10 | 11 | 12 | 13 |
|---|---|---|---|---|
| Count | 3 | 6 | 10 | 6 |

How many pupils are at most 11 years old? What frequency?

**5.** Running thread · Program → formula Procedure: multiply by 3 then subtract 2. Give $f(x)$ then $f(4)$.

**Sheet 46** Thu. 4 March 2027 Absolute value and square functions: sign and variations

**1.** Warm-up · Flash Calculate $|-5|$, $(-5)^2$ and $|2 - 7|$.

**2.** Core · Method Give the variation table of $x \mapsto |x|$ and its sign.

**3.** Core + · Reflection Compare $(-3)^2$ and $(-2)^2$ then $3^2$ and $2^2$. Is the square function increasing on $\mathbb{R}$ ?

**4.** Back · Method Two fair dice are rolled. Probability that the sum is 7?

**5.** Running thread · Method Procedure: choose a number; add 1; square it; subtract the square of the starting number. What is the result for 6? What general expression?

**Sheet 47** Mon. 8 March 2027 Reciprocal function: variations; relative position of y = x and y = $x^2$

**1.** Warm-up · Flash Calculate $\dfrac{1}{2}$, $\dfrac{1}{-4}$ and $\dfrac{1}{0{,}1}$.

**2.** Core · Method Compare $\dfrac{1}{3}$ and $\dfrac{1}{5}$; what do we say about the variation of the reciprocal function on $]0; +\infty[$ ?

**3.** Core + · True/false True or false? Justify. For every real number $x > 0$, $x^2 \geqslant x$.

**4.** Back · Flash What are the front view and the top view of a cylinder standing on its base?

**5.** Running thread · Prove Show that the product of two even integers is a multiple of 4.

**Sheet 48** Tue. 9 March 2027 Compare f $(a)$ *et* $f(b)$ ; optimisation problems

**1.** Warm-up · Flash $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.

**2.** Core · Method $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculation, using the variations.

**3.** Core + · Optimisation $A(x) = x(10 - x)$ for $x \in [0; 10]$ (area of a rectangle with perimeter 20). Show that $A(x) = 25 - (x - 5)^2$ and deduce the maximum.

**4.** Back · Flash 18 of 24 pupils have an average above 10. Give the frequency as a percentage.

**5.** Running thread · MCQ The solution of $\dfrac{x}{3} + 1 = \dfrac{7}{2}$ is:

**A.** $\dfrac{15}{2}$ **B.** $\dfrac{9}{2}$ **C.** $\dfrac{21}{2}$ **D.** $\dfrac{5}{6}$

## Cross-tabulating two categorical variables

**Sheet 49** Thu. 11 March 2027 Cross-tabulating two categorical variables

**1.** Warm-up · Flash Express $\frac{24}{60}$ as a percentage.

**2.** Core · Flash In a class of 30 pupils, 18 are girls. What is the frequency of girls? Of boys?

**3.** Core + · Classify Classify each variable: eye colour; baccalauréat grade (AB, B, TB); height in cm.

**4.** Back · Problem A rectangle with perimeter 20 cm has length $x$. Express its area $A(x)$; calculate $A(3)$ and $A(5)$.

**5.** Running thread · Flash Calculate the exact volume of a cylinder with radius 2 cm and height 5 cm.

**Sheet 50** Mon. 15 March 2027 Qualitative variables; two-way frequency table

**1.** Warm-up · Flash Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.

**2.** Core · Table Total 40 students. Girls: 22; half-boarders (DP): 25; girls DP: 15. Complete the two-way table.

| | DP | Non-boarders | Total |
|---|---|---|---|
| Girls | 15 | … | 22 |
| Boys | … | … | … |
| Total | 25 | … | 40 |

**3.** Core + · True/false True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.

**4.** Back · Problem Reduction A: $-30\%$ off 80,000 Ar. Discount B: $-20\%$ then $-10\%$. Which is the more advantageous?

**5.** Running thread · Method We know that $AB^2 + AC^2 = BC^2$. What conclusion do we draw? What property?

**Sheet 51** Tue. 16 March 2027 Marginal frequencies

**1.** Warm-up · Flash Express $\frac{25}{40}$ as a percentage.

**2.** Core · Method With the table from sheet 50: calculate the marginal frequency of girls and that of the half-boarders.

**3.** Core + · Flash The marginal frequency of girls is 55%. What is that of boys?

**4.** Back · Flash Calculate $(2a - b)^2$ for $a = 3$ and $b = -2$.

**5.** Running thread · Flash $A(-1;2)$ and $B(3;2)$. What can be said about the line $(AB)$ ? Give $AB$.

**Sheet 52** Thu. 18 March 2027 Conditional frequencies

**1.** Warm-up · Flash Give $\frac{15}{22}$ as a percentage rounded to one decimal place.

**2.** Core · Method With the same table: among the girls, what frequency of DP? Among the boys?

**3.** Core + · Distinguish Distinguish "the frequency of girls among DP pupils" from "the frequency of DP pupils among the girls".

**4.** Back · Construction steps We draw $[AB]$ of 6 cm, then the circles with centres $A$ and $B$ and radius 4 cm, which intersect at $C$ and $D$. Type of $ACBD$ ?

**5.** Running thread · Flash 91 prime?